

ILOMANTSI MEKRIJARVI, Finland

WMO: 029390

Lat: 62.7697N

Lon: 30.9753E

Elev: 513

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-12.5	-24.5	1.4	-18.9	-17.1	2.2	-12.3	20.8	30.3	18.7	29.3	2.4	30	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	79.6	65.7	76.2	64.1	72.9	62.2	68.4	76.2	66.2	73.1	64.1	70.8	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.6	72.5	63.4	89.2	70.1	61.4	83.0	67.7	32.9	76.0	31.1	73.2	29.6	70.8	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.6	15.4	13.6	-22.9	83.3	7.4	4.9	-28.2	86.9	-32.6	89.8	-36.7	92.6	-42.2	96.2		
			-23.1	70.9	7.3	3.0	-28.4	73.0	-32.7	74.7	-36.8	76.4	-42.1	78.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	14.0	15.0	23.1	35.0	48.3	55.8	62.4	58.8	49.7	37.8	29.4	22.4	(6)
(7)		DBStd	18.73	14.15	12.85	9.61	6.88	8.18	7.03	6.39	6.13	5.67	7.14	8.18	11.52	(7)
(8)		HDD50	5471	1116	980	835	451	129	22	2	6	74	381	619	856	(8)
(9)		HDD65	10015	1581	1400	1300	900	518	287	127	208	460	844	1069	1321	(9)
(10)		CDD50	1006	0	0	0	1	78	196	386	279	64	2	0	0	(10)
(11)		CDD65	74	0	0	0	0	1	11	46	16	0	0	0	0	(11)
(12)		CDH74	545	0	0	0	0	28	87	330	100	0	0	0	0	(12)
(13)		CDH80	81	0	0	0	0	0	10	58	13	0	0	0	0	(13)
(14)	Wind	WSAvg	6.3	5.7	6.2	6.7	6.5	6.7	6.0	5.3	5.4	6.0	6.8	7.3	6.9	(14)
(15)	Precipitation	PrecAvg	27.0	1.8	1.4	1.3	1.3	2.0	2.9	3.2	3.4	2.6	2.5	2.4	2.2	(15)
(16)		PrecMax	36.1	3.2	2.4	2.2	2.8	3.9	4.4	6.6	5.6	4.6	4.8	4.0	3.5	(16)
(17)		PrecMin	22.8	0.7	0.5	0.6	0.6	0.7	1.5	0.6	1.6	1.2	0.8	1.5	0.8	(17)
(18)		PrecStd	3.5	0.7	0.6	0.4	0.6	1.0	0.9	1.6	1.2	0.9	1.0	0.7	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	37.1	46.2	59.4	77.7	80.9	85.3	81.5	67.4	54.3	46.4	42.0	(19)
(20)			MCWB	35.2	34.9	38.7	46.8	58.5	64.2	69.3	67.5	60.0	52.5	45.4	41.1	(20)
(21)		2%	DB	34.2	34.3	40.6	53.8	72.6	76.3	80.9	76.5	63.8	51.2	43.3	38.1	(21)
(22)			MCWB	33.5	33.1	35.2	43.9	57.4	62.8	67.3	65.3	57.4	49.7	42.1	37.2	(22)
(23)		5%	DB	32.9	33.0	37.2	49.6	67.2	72.4	77.6	72.8	60.7	49.3	41.3	36.0	(23)
(24)			MCWB	32.4	32.2	33.6	41.0	54.6	60.2	66.0	63.1	56.1	47.9	40.5	35.2	(24)
(25)		10%	DB	31.8	31.6	34.7	45.8	62.8	68.6	74.3	69.4	58.2	47.3	39.1	34.3	(25)
(26)			MCWB	31.4	31.1	32.4	38.9	51.5	58.3	64.1	61.6	54.9	45.9	38.5	33.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.7	35.1	39.3	49.4	62.8	68.3	71.7	69.7	62.0	53.0	45.3	41.1	(27)
(28)			MCD B	36.2	36.0	45.7	55.6	72.2	76.4	80.8	78.8	65.5	54.0	46.0	41.8	(28)
(29)		2%	WB	33.8	33.6	36.1	44.9	58.9	64.9	69.4	66.7	59.2	50.3	42.2	37.5	(29)
(30)			MCD B	34.2	34.3	39.4	52.7	69.2	73.0	78.2	73.3	62.1	50.9	43.0	38.0	(30)
(31)		5%	WB	32.5	32.4	34.2	41.9	56.2	62.0	67.5	64.7	57.1	48.2	40.6	35.3	(31)
(32)			MCD B	32.8	33.0	36.1	48.0	65.2	69.3	75.1	71.0	59.9	48.9	41.3	35.8	(32)
(33)		10%	WB	31.4	31.0	33.0	39.6	53.1	59.5	65.5	62.8	55.2	46.0	38.5	33.8	(33)
(34)			MCD B	31.8	31.6	34.5	44.5	60.9	67.2	72.3	68.2	57.8	47.1	39.1	34.2	(34)
(35)	Mean Daily Temperature Range	MDBR	10.0	11.7	14.9	14.7	16.8	16.0	15.9	14.2	11.4	7.6	6.2	8.3	(35)	
(36)		5% DB	MCD BR	5.6	7.9	15.0	20.7	25.3	21.2	20.3	19.8	15.7	9.0	6.7	6.2	(36)
(37)			MCWBR	5.6	7.8	10.8	12.5	12.7	10.5	9.3	9.9	10.2	7.3	6.5	5.9	(37)
(38)		5% WB	MCD BR	5.6	7.8	11.4	17.5	21.7	18.0	18.6	16.5	13.1	8.2	6.6	5.7	(38)
(39)			MCWBR	5.6	7.8	8.9	11.3	12.1	9.8	9.4	9.1	9.3	7.1	6.5	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.231	0.284	0.310	0.347	0.351	0.344	0.372	0.353	0.332	0.310	0.246	0.209	(40)	
(41)		taud	2.005	2.106	2.158	2.277	2.443	2.488	2.425	2.460	2.509	2.399	2.096	1.963	(41)	
(42)		Ebn at Noon	128	204	248	265	275	279	266	262	246	201	121	71	(42)	
(43)		Edh at Noon	11	22	30	33	31	30	31	28	22	16	10	6	(43)	
(44)	All-Sky Solar Radiation	RadAvg	62	242	701	1165	1530	1662	1573	1195	663	253	63	28	(44)	
(45)		RadStd	8	55	90	122	168	107	153	130	81	36	11	3	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (24 neighbors)	+1.06	N/A	N/A	N/A	N/A	-373	-365	N/A	N/A	(47)

Nomenclature: See separate page